

KARLSTAD SOL, Sweden

WMO: 024150

Lat: 59.3592N

Lon: 13.4719E

Elev: 46

StdP: 100.77

Time Zone: 1.00 (EUC)

Period: 98-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.8	-14.9	-19.8	0.7	-17.7	-16.9	0.9	-13.9	10.6	0.9	9.6	0.4	2.1	350	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	26.2	17.1	24.8	17.0	22.9	16.1	18.8	23.4	18.0	22.1	17.2	21.1	3.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.3	19.9	16.2	11.6	19.3	15.8	11.3	18.9	53.5	23.3	50.7	22.3	48.4	21.1	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	7.1	DB	-18.9	29.2	4.6	1.9	-22.2	30.6	-24.8	31.6	-27.4	32.7	-30.7	34.0	(4)
(5)			WB	-19.7	20.4	4.3	0.6	-22.8	20.9	-25.3	21.2	-27.7	21.6	-30.8	22.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-2.8	-3.2	-0.1	5.8	10.2	14.2	16.9	15.7	11.8	6.1	1.8	-2.3	(6)	
	DBStd	8.17	5.39	4.74	4.04	3.14	3.16	2.75	2.60	2.67	2.92	3.68	4.20	6.02	(7)	
	HDD10.0	2017	396	369	312	131	35	1	0	1	17	127	246	383	(8)	
	HDD18.3	4446	654	603	571	375	253	128	62	88	197	378	496	641	(9)	
	CDD10.0	641	0	0	0	5	41	128	213	177	69	7	0	0	(10)	
	CDD18.3	27	0	0	0	0	0	4	17	7	0	0	0	0	(11)	
	CDH23.3	288	0	0	0	0	3	70	159	56	0	0	0	0	(12)	
	CDH26.7	36	0	0	0	0	0	10	24	2	0	0	0	0	(13)	
(14) Wind	WSAvg	3.4	3.7	3.4	3.4	3.3	3.5	3.3	3.1	3.1	3.4	3.4	3.6	3.4	(14)	
(15) Precipitation	PrecAvg	640	47	33	36	40	46	57	61	77	66	70	71	49	(15)	
	PrecMax	871	131	93	97	108	106	186	185	156	136	176	162	89	(16)	
	PrecMin	452	2	1	3	0	9	7	4	1	14	14	16	6	(17)	
	PrecStd	112	29	20	21	23	25	37	33	40	31	38	36	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	7.8	12.8	19.8	23.2	28.1	29.1	27.0	22.2	15.1	11.2	8.9	(19)	
		MCWB	6.4	5.7	6.9	12.3	13.3	17.0	18.5	17.6	16.4	12.2	9.5	8.6	(20)	
	2%	DB	6.1	5.9	9.8	16.8	21.2	25.2	27.0	25.1	20.1	13.8	10.1	7.8	(21)	
		MCWB	5.0	4.2	5.6	10.4	12.8	16.0	17.7	17.5	14.8	11.9	8.9	7.4	(22)	
	5%	DB	5.0	4.2	7.0	14.2	19.1	22.8	25.1	23.2	18.2	12.8	9.0	6.8	(23)	
		MCWB	3.9	3.0	3.8	8.9	11.7	15.0	17.3	17.1	13.9	11.1	8.2	6.3	(24)	
	10%	DB	3.9	3.1	5.2	12.2	16.9	20.8	23.1	21.1	16.9	11.2	7.9	5.2	(25)	
		MCWB	3.2	2.1	3.1	7.4	10.8	14.3	16.7	16.4	13.7	9.9	7.3	4.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	5.8	7.1	13.0	15.5	18.4	20.3	20.1	17.3	13.5	10.4	8.3	(27)	
		MCDB	7.4	6.3	11.0	16.6	19.4	24.0	25.7	23.6	20.8	14.5	11.2	8.4	(28)	
	2%	WB	5.2	4.3	5.8	10.5	13.8	17.1	19.3	18.6	15.8	12.3	9.2	7.1	(29)	
		MCDB	5.7	5.1	8.6	15.1	19.1	22.8	24.4	22.4	18.7	13.5	9.7	7.2	(30)	
	5%	WB	3.9	3.2	4.6	9.5	12.7	16.0	18.4	17.7	15.0	11.3	8.0	5.9	(31)	
		MCDB	4.4	4.0	6.2	13.8	17.5	21.3	23.0	21.3	17.4	12.5	8.7	6.2	(32)	
	10%	WB	2.9	2.1	3.5	8.2	11.5	15.0	17.5	16.9	14.0	10.5	7.0	4.7	(33)	
		MCDB	3.5	3.0	4.6	11.1	16.1	19.6	21.6	20.3	16.4	11.5	7.5	5.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.8	7.5	9.7	10.0	10.1	9.2	9.0	8.6	6.5	4.8	5.1	(35)	
		MCDBR	5.3	5.9	10.4	13.2	14.1	14.1	12.8	11.8	10.5	6.9	5.8	4.5	(36)	
	5% WB	MCWBR	5.2	5.0	7.1	7.7	7.0	6.6	5.8	5.9	6.4	5.0	5.4	4.7	(37)	
		MCWBR	4.9	5.4	8.7	11.9	11.7	11.9	10.9	9.7	8.9	6.1	5.8	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.263	0.304	0.328	0.366	0.363	0.354	0.375	0.367	0.338	0.327	0.288	0.243	(40)	
		taud	2.211	2.298	2.388	2.362	2.432	2.481	2.440	2.456	2.523	2.467	2.301	2.135	(41)	
	Edn at Noon		545	712	812	837	867	881	850	830	801	687	512	421	(42)	
		Edh at Noon	43	67	84	102	102	99	101	92	74	58	40	31	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.31	0.88	2.43	3.83	4.99	5.75	5.13	4.03	2.69	1.20	0.40	0.21	(44)	
	RadStd		0.06	0.17	0.24	0.51	0.47	0.34	0.64	0.40	0.32	0.22	0.09	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page